

The Mosaic of Lin

On a hill above Lake Ohrid, the village of Lin preserves mosaic floors from a sixth-century basilica. Discovered in 1967, the mosaics feature geometric, plant, and animal motifs.

Conservators are preparing a digital reconstruction of one damaged geometric panel. The panel is represented by an $N \times M$ matrix A of non-negative integer intensity values. The reconstruction may restore faded detail by increasing an intensity, but it must not erase recorded detail by decreasing one.

A matrix B is **bitonic** if there is a **peak** cell (p, q) such that every row rises towards column q and falls after it, and every column rises towards row p and falls after it. Precisely:

- $B[i][1] \leq \dots \leq B[i][q] \geq \dots \geq B[i][M]$ for every row i .
- $B[1][j] \leq \dots \leq B[p][j] \geq \dots \geq B[N][j]$ for every column j ;

All inequalities are non-strict, so adjacent values may be equal. Border peaks are allowed. If $p = 1$, every column is non-increasing; if $p = N$, every column is non-decreasing. Similarly, $q = 1$ means every row is non-increasing, while $q = M$ means every row is non-decreasing.

For every cell, choose an integer $B[i][j] \geq A[i][j]$. The restoration cost of a cell is $B[i][j] - A[i][j]$, and the total restoration cost is the sum of the costs of all cells.

Find a bitonic reconstruction of minimum total cost.

Input format

The first line contains two integers N and M .

Each of the next N lines contains M integers: the matrix A .

Output format

On the first line, print the minimum total cost.

On the second line, print the one-based coordinates p and q of a peak of your matrix.

On each of the next N lines, print M integers: the matrix B .

Any optimal matrix, together with a peak that it is bitonic for, is accepted.

Constraints

- $1 \leq N, M \leq 2000$
- $0 \leq A[i][j] \leq 10^9$

Subtasks

Subtask	Points	Additional constraints
1	7	$N = M = 1$
2	13	$N = 1$ or $M = 1$
3	20	$N, M \leq 30$
4	25	$N, M \leq 300$
5	35	No additional restrictions.

Examples

Example 1

Input:

```
3 3
1 8 2
5 3 4
2 6 1
```

Output:

```
5
2 2
1 8 2
5 8 4
2 6 1
```

Only the centre cell is raised from 3 to 8, so the cost is 5.

Note that the peak is the centre cell, which initially had intensity 3, and **not** the cell that initially had the largest intensity, 8. Choosing the cell with the largest initial intensity as the peak is not optimal here: the best reconstruction with the peak at $(1, 2)$ costs 9.

Example 2

Input:

```
1 7
7 1 9 2 6 3 5
```

Output:

```
12
1 3
7 7 9 6 6 5 5
```

A single row must simply rise to column q and fall after it. With $q = 3$, the three raised cells cost $6 + 4 + 2$, and no other choice of q is cheaper.

Example 3

Input:

```
2 3
5 1 5
1 1 1
```

Output:

```
4
1 1
5 5 5
1 1 1
```

The peak lies in the top-left corner, so every row and every column is non-increasing. The cell with intensity 1 in the top row must be raised to 5, because it lies between two cells with intensity 5 and the top row must be non-increasing.